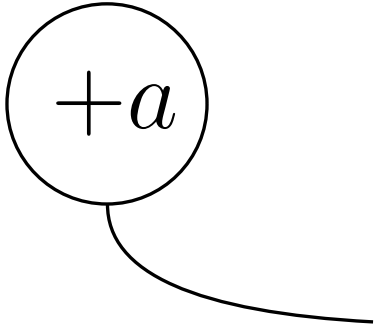
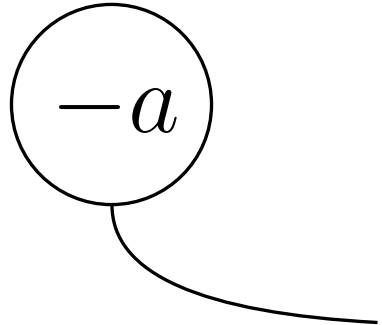
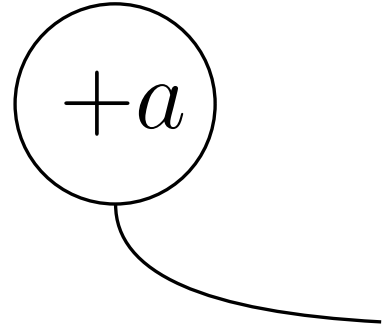


with $0 < b \leq a$



with $0 < a < b$



with $0 < a$